

Exercise sheet 1

PRISM - Curves and Surfaces

1. Can you find a parametrization of the circle of radius r which has speed 1? What is its acceleration?
2. Consider a line given as the set of points that satisfy the equation $y = mx + b$. Find a parametrization of it which has speed 1. What is its acceleration?
3. Consider a parametrization $f : (a, b) \rightarrow \mathbb{R}^2$. Assume that its acceleration is 0. Show that it must be a straight line.
4. Suppose you are given a parametrization which has constant speed, but the speed is not 1. How can you make the speed 1?